Notes on open problems 2 and 3 of *Prime extremality for product integrals over primitive sets*

Working notes (draft), 31 October 2026

Abstract

These notes continue the paper [1], whose notation and numbering they use. For open problem 3 we show that, for *every* N, the truncation error $I_N - \Lambda$ equals, up to a factor $1 + o(1)$, an explicit smooth factor times the Laplace transform $\Sigma_q = \sum_{p \geq q} e^{-(p-q)t_q}$ of the primes after the least prime $q > N$, at scale $1/t_q \sim \sqrt{q \log q/(2\zeta(2))}$. Consequences: two-sided bounds for every N that sharpen Proposition 5.5; the conjectured formula of open problem 3 holds for all $N \leq X$ outside a set of size $O_A(X(\log X)^{-A})$; and if it held for all N, then $p_{n+1} - p_n \ll \sqrt{p_n \log p_n}$, a bound not known even under the Riemann hypothesis. Numerically, the local factor explains the residual of open problem 3 almost exactly (for $20000 < N \leq 10^5$: correlation 0.9998, fitted slope 1.001). For open problem 2 we determine the *relative* cone exactly: $\mathrm{Min}(U)$ minimizes $\mathbb{E}\phi(Z_A)$ over primitive $A \subseteq U$ for every U if and only if $\mathbb{E}\phi(Z_{\mathcal{P}}^{1/D}) \geq \mathbb{E}\phi(V^{1/D})$ for every integer $D \geq 2$, with V uniform on $[0,1]$; for $\phi(x) = x^k$ this means $k \leq 2\alpha_0$, so the barrier of Remark 4.4 is intrinsic to the relative problem. For the absolute problem, compressing at odd primes before the prime 2 removes that barrier: the primes minimize $\mathbb{E}\min(Z_A^k, c^k)$ for all $0 < k \leq 3b$ whenever $J_b(1) > 0$. This gives $k \leq 51/8$, formally verified in Lean, and with exact envelope certificates every $0 < k < 3 + 6\sqrt{3} = 13.39\ldots$. Keeping the weight contributed by the large primes, and comparing the $\{2,3,5,7\}$-smooth part with $\{2,3,5,7\}$ in one step, we determine the exact threshold: the primes minimize $\mathbb{E}\min(Z_A^k, c^k)$ for every N if and only if $k \leq k_9 = 125.8404540694\ldots$, the exponent at which $A_9 = \{4,5,6,7,9\}$ ties with $\mathcal{P}_9$. The remaining finitely many cases are verified by 5,350 certificates in exact arithmetic. Theorems 3.2, 3.5, 3.7 and 3.8 are formalized in Lean (§4).

1 Summary

Recall from [1]: $P_A(x) = \prod_{n \in A}(1 - x^n)$, $I_N = \int_0^1 P_{\mathcal{P}_N}$, $\Lambda = \lim I_N$, $\Phi(t) = -\sum_p \log(1 - e^{-pt})$, $a = \zeta(2)$, $L(t) = \log(1/t)$, $M_N = \inf_{t>0}(Nt + \Phi(t))$, attained at the unique t_N with $-\Phi'(t_N) = N$, and q_N the least prime exceeding N. Open problem 3 asks for the term after M_{q_N} in $-\log(I_N - \Lambda)$; the paper's heuristic is

$$-\log(I_N - \Lambda) = M_q + \tfrac{1}{4}\log q + \tfrac{3}{4}\log\log q - \tfrac{1}{2}\log\pi + \tfrac{1}{4}\log(2\zeta(2)) + o(1), \qquad q = q_N, \quad (1)$$

and the paper asks whether (1) holds for almost all N.

- **Theorem 2.4** (local law, every N): $I_N - \Lambda = (1 + O(q^{-1/4}\log^4 q))\sqrt{2\pi/\Phi''(t_q)}\, e^{-M_q - t_q}\, \Sigma_q$ with $\Sigma_q = \sum_{p \geq q} e^{-(p-q)t_q}$.
- **Corollary 2.5**: for every $N \geq 3$, $M_q + \frac{1}{4}\log q + \frac{3}{4}\log\log q - C \leq -\log(I_N - \Lambda) \leq M_q + \frac{3}{4}\log q + \frac{1}{4}\log\log q + C$.

- **Theorem 2.7**: (1) holds for all N outside a set $\mathcal{E}$ with $\#(\mathcal{E}\cap[1,X])\ll_A X(\log X)^{-A}$. This answers the question of open problem 3.
- **Corollary 2.9**: if (1) holds for all large N, then $p_{n+1}-p_n\le(\log 4/\sqrt{2\zeta(2)}+o(1))\sqrt{p_n\log p_n}$.
- **Theorem 3.2**: the exact cone for the relative version of open problem 2; **Corollary 3.3**: for powers, relative extremality holds iff $k\le 2\alpha_0$.
- **Theorem 3.5**: absolute moment minimality for $0<k\le 3b$ whenever $J_b(1)>0$; with $b=17/8$ this is $k\le 51/8$, formalized (`RntPaper/MomentsOdd.lean`). **Theorem 3.7**: every $0<k<3+6\sqrt{3}$.
- **Theorem 3.8**: the largest admissible power is exactly $k_9=125.84045406944565748\ldots$ (computer-assisted, exact arithmetic).

2 The truncation error

Throughout, p and q denote primes. For an integer $n\ge 2$ put $P_{<n}=\prod_{p<n}(1-x^p)$ and

$$\delta(n)=\int_0^1 x^n P_{<n}(x)\,dx.$$

For a prime p, $I_{p-1}-I_p=\delta(p)$, so $I_N-\Lambda=\sum_{p>N}\delta(p)=\sum_{p\ge q_N}\delta(p)$.

2.1 The saddle

Lemma 2.1 (derivatives of Φ). *For $j=0,1,2,3$, $(-1)^j t^{j+1}L(t)\,\Phi^{(j)}(t)\to j!\,a$ as $t\downarrow 0$.*

Proof. $j=0$ is [P, Lemma 5.2]. Let $j\ge 1$ and $S_j(u)=\sum_p p^j e^{-pu}$. Partial summation and $\theta(x)=x+o(x)$ give $u^{j+1}\sum_p p^j\log p\,e^{-pu}\to\int_0^\infty y^j e^{-y}dy=j!$, and the splitting argument of [P, Lemma 5.1] at $u^{-(1\pm\delta)}$ turns this into $u^{j+1}L(u)S_j(u)\to j!$. Differentiating $\Phi(t)=\sum_{m\ge1}S(mt)/m$ termwise (all series of derivatives converge locally uniformly on $(0,\infty)$),

$$(-1)^j t^{j+1}L(t)\Phi^{(j)}(t)=\sum_{m\ge1}\frac{1}{m^2}\,(mt)^{j+1}L(mt)S_j(mt)\,\frac{L(t)}{L(mt)}.$$

Each term tends to $j!/m^2$. Domination is as in the proof of [P, Lemma 5.2], using $S_j(u)\le\sum_n n^j e^{-nu}\ll_j u^{-j-1}e^{-u/2}$ and $u^{j+1}L(u)S_j(u)\le 2\,j!$ for small u. □

Lemma 2.2 (the saddle). *As $n\to\infty$ through real numbers, $t_n\sim\sqrt{2a/(n\log n)}$, $L(t_n)\sim\frac12\log n$, $\Phi''(t_n)\sim 2n/t_n$, and $dt_n/dn=-1/\Phi''(t_n)\sim -t_n/(2n)$. Moreover $|\Phi'''(t)|\ll n/t_n^2$ for $t_n/2\le t\le 2t_n$, and $M_n\asymp\sqrt{n/\log n}$.*

Proof. By Lemma 2.1, $n=-\Phi'(t_n)\sim a/(t_n^2L(t_n))$; taking logarithms gives $L(t_n)\sim\frac12\log n$ and then t_n. The other statements follow from Lemma 2.1, since $L(t)\sim L(t_n)$ uniformly on $[t_n/2,2t_n]$; differentiate $-\Phi'(t_n)=n$ for dt_n/dn. For M_n see the proof of [P, Theorem 1.5]. □

Proposition 2.3 (Laplace). *As $n\to\infty$,*

$$\delta(n)=\sqrt{\frac{2\pi}{\Phi''(t_n)}}\;e^{-M_n-t_n}\,\bigl(1+O(n^{-1/4}\log^4 n)\bigr).$$

Proof. With $x = e^{-t}$, $\delta(n) = \int_0^\infty e^{-(n+1)t} P_{<n}(e^{-t})\,dt$ and $P_{<n}(e^{-t}) = e^{-\Phi(t)} R_n(t)$, where $R_n(t) = \prod_{p \ge n}(1-e^{-pt})^{-1} \ge 1$ satisfies $\log R_n(t) \le e^{-nt}/((1-e^{-nt})(1-e^{-t}))$. Write $F(t) = nt + \Phi(t)$, which is strictly convex with minimum M_n at t_n. Put $\sigma = \Phi''(t_n)^{-1/2}$, $K = \log n$ and $\rho = K\sigma$. By Lemma 2.2, $\sigma/t_n \asymp (nt_n)^{-1/2}$, which is a negative power of n, so $\rho \le t_n/2$ for large n.

The window $|t - t_n| \le \rho$. By Taylor's theorem, $F(t) = M_n + \frac{1}{2}\Phi''(t_n)(t-t_n)^2 + r(t)$ with $|r(t)| \ll (n/t_n^2)\rho^3 \asymp K^3(nt_n)^{-1/2} \to 0$. Here $t \ge t_n/2$, so $e^{-t} = e^{-t_n}(1+O(\rho))$ and $\log R_n(t) \ll e^{-nt_n/2}/t_n$, which is smaller than any power of n. The window therefore contributes $e^{-M_n - t_n}\sqrt{2\pi}\,\sigma\,(1 + O(K^3(nt_n)^{-1/2}))$, the Gaussian mass outside K standard deviations being $O(e^{-K^2/2})$; note $K^3(nt_n)^{-1/2} \ll n^{-1/4}\log^4 n$.

The range $t \ge s_0 := 2\log n/n$ *outside the window.* There $e^{-nt} \le n^{-2}$ and $1 - e^{-t} \ge s_0/2$, so $R_n(t) \le e$. Convexity gives $F(t) - M_n \ge \frac{|t-t_n|}{\rho}\min_\pm\big(F(t_n \pm \rho) - M_n\big) \ge \frac{K^2}{3}\frac{|t-t_n|}{\rho}$ for $|t - t_n| \ge \rho$, so this range contributes at most $2e\,e^{-M_n}\int_\rho^\infty e^{-K^2 h/(3\rho)} dh = O\big(\sigma K^{-1} e^{-K^2/3} e^{-M_n}\big)$.

The range $0 < t < s_0$. Here $P_{<n}(e^{-t}) \le P_{<n}(e^{-s_0}) \le e^{1-\Phi(s_0)}$. By [P, Lemma 5.2], $\Phi(s_0) \asymp n/\log^2 n$, which exceeds $2M_n \asymp \sqrt{n/\log n}$ for large n. The contribution is at most $s_0 e^{1-2M_n} = o(\sigma e^{-M_n})$, because σ is a power of n. □

The factor $e^{-t_n} = 1 + O(t_n)$ could be absorbed in the error term; we keep it because it is the Jacobian of $x = e^{-t}$ and it is most of the numerical discrepancy. Numerically, $\log\big(\delta(p)/\sqrt{2\pi/\Phi''(t_p)}\,e^{-M_p - t_p}\big)$ is 0.0211, 0.0073, 0.0029, 0.0011 at $p \approx 10^2, 10^3, 10^4, 10^5$.

2.2 The local law

Theorem 2.4 (local law)**.** *For every $N \ge 2$, with $q = q_N$,*

$$I_N - \Lambda = \big(1 + O(q^{-1/4}\log^4 q)\big)\sqrt{\frac{2\pi}{\Phi''(t_q)}}\,e^{-M_q - t_q}\,\Sigma_q, \qquad \Sigma_q := \sum_{p \ge q} e^{-(p-q)t_q}.$$

Proof. Put $\Lambda_u = \sqrt{2\pi/\Phi''(t_u)}\,e^{-M_u - t_u}$ for real $u \ge 2$. By Proposition 2.3, $\delta(p) = \Lambda_p(1 + O(q^{-1/4}\log^4 q))$ for all $p \ge q$. On the range in (a), $e^{t_q - t_p} = 1 + O(Ht_q/q)$, and below these factors are bounded by 1. We write $o(1)$ for errors that are $O(q^{-1/4}\log^4 q)$. Let $H = 3\log q/t_q$.

(a) Let $q \le p \le q + H$. Since $M_u' = t_u$ and $|dt_u/du| \le (1 + o(1))t_q/q$ on $[q, q+H]$ (Lemma 2.2),

$$\big|M_p - M_q - (p-q)t_q\big| \le (1+o(1))H^2 t_q/q = O\Big(\frac{\log^2 q}{qt_q}\Big) = o(1),$$

because $qt_q \asymp \sqrt{q/\log q}$. Also $\Phi''(t_p)/\Phi''(t_q) \to 1$ uniformly, since $\Phi''(t_u) \sim 2u/t_u$ and $H = o(q)$. Hence $\Lambda_p = \Lambda_q e^{-(p-q)t_q}(1 + o(1))$ uniformly in this range.

(b) Let $q + H < p \le 2q$. Then $M_p - M_q \ge (p-q)t_{2q} \ge (p-q)t_q/2$, since $t_{2q}/t_q \to 2^{-1/2}$, and $\Phi''(t_p) \ge (1+o(1))\Phi''(t_q)$, so $\sum_{q+H<p\le 2q}\Lambda_p \le 2\Lambda_q \sum_{m>H} e^{-mt_q/2} \le \Lambda_q \cdot 8q^{-3/2}/t_q = o(\Lambda_q)$. For $p > 2q$, $M_p - M_q \ge \frac{p}{2}t_p \gg \sqrt{p/\log p}$ while $\Lambda_q \gg e^{-M_q}q^{-1}$, so $\sum_{p>2q}\Lambda_p = o(\Lambda_q)$.

(c) Similarly $\sum_{p>q+H} e^{-(p-q)t_q} \le e^{-Ht_q}/(1-e^{-t_q}) = O(q^{-3}/t_q) = o(1)$.

Combining (a)–(c), $\sum_{p\ge q}\delta(p) = \Lambda_q\big[(1+o(1))\sum_{q\le p\le q+H} e^{-(p-q)t_q} + o(1)\big] = \Lambda_q\Sigma_q(1+o(1))$, since $\Sigma_q \ge 1$ (the term $p = q$). All errors in (a)–(c) are $O(q^{-1/4}\log^4 q)$: the linearization costs $O(\log^2 q/(qt_q)) = O(q^{-1/2}\log^{5/2} q)$, and the tails are smaller than any power of q. □

2.3 Every N

Corollary 2.5. *For every $N \geq 3$, with $q = q_N$,*

$$-\log(I_N - \Lambda) = M_q + t_q + \tfrac{1}{2}\log\frac{\Phi''(t_q)}{2\pi} - \log\Sigma_q + o(1), \qquad 1 \leq \Sigma_q \leq \frac{4 + o(1)}{t_q \log q},$$

and consequently, for an absolute constant C,

$$M_q + \tfrac{1}{4}\log q + \tfrac{3}{4}\log\log q - C \;\leq\; -\log(I_N - \Lambda) \;\leq\; M_q + \tfrac{3}{4}\log q + \tfrac{1}{4}\log\log q + C.$$

Proof. The identity is Theorem 2.4. For the upper bound on Σ_q write $\Sigma_q = t_q \int_0^\infty e^{-ht_q}\,\pi_q(h)\,dh$, where $\pi_q(h) = \#\{p \in [q, q+h]\}$, and use the Brun–Titchmarsh inequality $\pi(x+y) - \pi(x) \leq 2y/\log y$ $(y > 1)$ of Montgomery and Vaughan [6]: $\Sigma_q \leq (2 + o(1))/(t_q \log(1/t_q))$ and $\log(1/t_q) \sim \tfrac{1}{2}\log q$. The last display follows from $\Phi''(t_q) \sim 2q/t_q$ and $t_q \sim \sqrt{2a/(q\log q)}$. □

Asymptotically this sharpens [P, Proposition 5.5], whose window is $[M_q - 2\log q - 3,\ M_q + \log(q+1) + 4]$; that proposition, however, is explicit for every N and does not use the prime number theorem, whereas C here is not explicit. The lower bound is attained, up to $O(1)$, for almost all N (Theorem 2.7). The upper bound would require $\Sigma_q = O(1)$, that is a prime gap after q of size $\gg \log q/t_q \asymp \sqrt{q}(\log q)^{3/2}$, the scale of [P, Corollary 5.6(iii)]; whether such gaps occur is open.

2.4 Almost every N

The input is the prime number theorem in almost all short intervals, a classical consequence of Huxley's zero-density estimate [3]. We use it in the following form; see [4, Theorem 7] for a proof, and [5] for the exponent $1/5$, which also suffices here.

Lemma 2.6 (Huxley)**.** *Fix $\varepsilon, A > 0$. For $X \geq X_0(\varepsilon, A)$ and $X^{-5/6+\varepsilon} \leq \delta \leq 1$, the set of $x \in [X, 2X]$ with $|\psi(x) - \psi(x - \delta x) - \delta x| \geq \delta x/\log x$ has measure $O_A(X(\log X)^{-A})$.*

Theorem 2.7. *For every $A > 0$ there is a set $\mathcal{E}_A \subseteq \mathbb{N}$ with $\#(\mathcal{E}_A \cap [1, X]) \ll_A X(\log X)^{-A}$ such that, as $N \to \infty$ with $N \notin \mathcal{E}_A$,*

$$t_q \log q\, \Sigma_q = 1 + O\Big(\frac{1}{\log N}\Big), \qquad q = q_N.$$

Consequently (1) holds for all $N \notin \mathcal{E}_A$, in particular for almost all N.

Proof. Let X be large and $N \in (X, 2X]$. Put $H = \sqrt{X\log X}$, $J = \lceil(\log X)^4\rceil$ and $\delta_j = jH/(X(\log X)^2)$ for $1 \leq j \leq J$. Each δ_j lies in the range of Lemma 2.6. For each j let $E_j \subseteq [X, 4X]$ be the exceptional set of the lemma for δ_j, applied with $A+5$ in place of A on $[X, 2X]$ and on $[2X, 4X]$; it has measure $O(X(\log X)^{-A-5})$. For real $x \in (X, 2X+1]$ put $y_j(x) = x/(1-\delta_j) \in (X, 4X)$, so that $(y_j(1-\delta_j), y_j] = (x, x + h_j(x)]$ with $h_j(x) = \delta_j x/(1-\delta_j)$. The set of x for which $y_j(x) \in E_j$ for some j has measure at most $\sum_j \mathrm{meas}(E_j) = O(X(\log X)^{-A-1})$. Call an integer N *good* if some $x \in [N, N+1)$ avoids it; all but $O(X(\log X)^{-A})$ integers in $(X, 2X]$ are good.

Let N be good. In windows of length $h \le H(\log X)^2$, prime powers p^m with $m \ge 2$ contribute $O((\log X)^4)$ to ψ, and $\log p = \log N + O(h/N)$; moving from x to N changes each count by $O(1)$. So for all j

$$\pi(N+h_j) - \pi(N) = \frac{h_j}{\log N}\Big(1 + O\Big(\frac{1}{\log X}\Big)\Big) + O(1), \qquad h_j = h_j(N).$$

The h_j form a grid of mesh $\asymp H/(\log X)^2$ covering $[(N/X)H/(\log X)^2, (N/X)H(\log X)^2] \supseteq [H/\log X, H\log X]$, so by monotonicity $\pi(N+h) - \pi(N) = (h/\log N)(1 + O(1/\log X))$ for all $h \in [H/\log X, H\log X]$. In particular $q - N \le h_1 = O(H/(\log X)^2)$, and the same estimate holds for $\pi_q(h) = \#\{p \in [q, q+h]\}$.

Since $t_q H$ lies between two positive constants (numerically in $[1.28, 1.81]$), write $\Sigma_q = \int_0^\infty e^{-v}\pi_q(v/t_q)\,dv$ and split at $v = c/\log X$ and $v = C\log X$ with $c \ge t_q H$ and $C \le t_q H$ (for instance $c = 2$, $C = 1$), so that the middle range has $v/t_q \in [H/\log X, H\log X]$. The middle range gives $(1 + O(1/\log X))/(t_q \log q)$. On $v < c/\log X$, Brun–Titchmarsh bounds the contribution by $O\big((t_q\log q)^{-1}(\log X)^{-2}\big) + O(1/\log X)$; on $v > C\log X$ it is $O\big((t_q\log q)^{-1}X^{-1}\big)$. Hence $t_q \log q\, \Sigma_q = 1 + O(1/\log X)$.

Finally, Corollary 2.5 together with $t_q \to 0$, $\Phi''(t_q) \sim 2q/t_q$ and $t_q \sim \sqrt{2a/(q\log q)}$ gives $t_q + \frac12\log\frac{\Phi''(t_q)}{2\pi} + \log(t_q\log q) = \frac14\log q + \frac34\log\log q - \frac12\log\pi + \frac14\log(2a) + o(1)$, which is (1). Summing over dyadic ranges bounds $\#(\mathcal{E}_A \cap [1, X])$. □

Remark 2.8. For $N \notin \mathcal{E}_A$ the $o(1)$ in (1) has three sources: the error $O(q^{-1/4}\log^4 q)$ of Theorem 2.4, the $O(1/\log N)$ in $t_q \log q\, \Sigma_q$, and the replacement of $\Phi''(t_q)$ and t_q by their first-order asymptotics. The last one dominates; second-order expansions of Φ' and Φ'', analogous to Step 2 of [P, Theorem 5.10], should make it $O(\log\log q/\log q)$, but we have not written this out. Numerically it is the mean offset ≈ -0.058 of the residual near $N = 10^5$ (Figure 1). The intrinsic form $-\log(I_N - \Lambda) = M_q + t_q + \frac12\log\frac{\Phi''(t_q)}{2\pi} - \log\Sigma_q$ has error in $[0.0001, 0.0012]$ for $20000 < N < 10^5$.

2.5 All N

Corollary 2.9. *If* (1) *holds for all large* N, *then* $p_{n+1} - p_n \le \big(\log 4/\sqrt{2\zeta(2)} + o(1)\big)\sqrt{p_n \log p_n}$.

Proof. Let q be a large prime, g the gap after it, and $N = q - 1$, so $q_N = q$. By Theorem 2.4, (1) at N means $t_q \log q\, \Sigma_q \to 1$. Brun–Titchmarsh applied to the primes beyond $q + g$ gives $\Sigma_q \le 1 + e^{-gt_q}\big(1 + (2 + o(1))/(t_q\log(1/t_q))\big)$, hence $t_q\log q\,\Sigma_q \le o(1) + (4 + o(1))e^{-gt_q}$. So $e^{-gt_q} \ge \frac14 - o(1)$, that is $g \le (\log 4 + o(1))/t_q$. □

The Riemann hypothesis gives only $p_{n+1} - p_n \ll \sqrt{p_n}\log p_n$ [2]. So (1) for every N lies beyond the Riemann hypothesis. It would follow from the expected asymptotic $\pi(x+h) - \pi(x) \sim h/\log x$ uniformly for $\sqrt{x}/\log x \le h \le \sqrt{x}\log^2 x$.

2.6 Numerics

Figure 1 and Table 1 use the exact telescoping terms $\delta(p)$ for $p < 1.6 \cdot 10^5$ and $N = p_n \le 10^5$, computed in floating point as in `next_prime_check.py` (`op3_local.py`). The residual of (1) lies in $[-0.39, 0.31]$ (in $[-0.25, 0.16]$ for $N > 20000$, standard deviation 0.057). The residual of Theorem 2.4 lies in $[0.0001, 0.0012]$ for $N > 20000$. Regressing the residual of (1) on

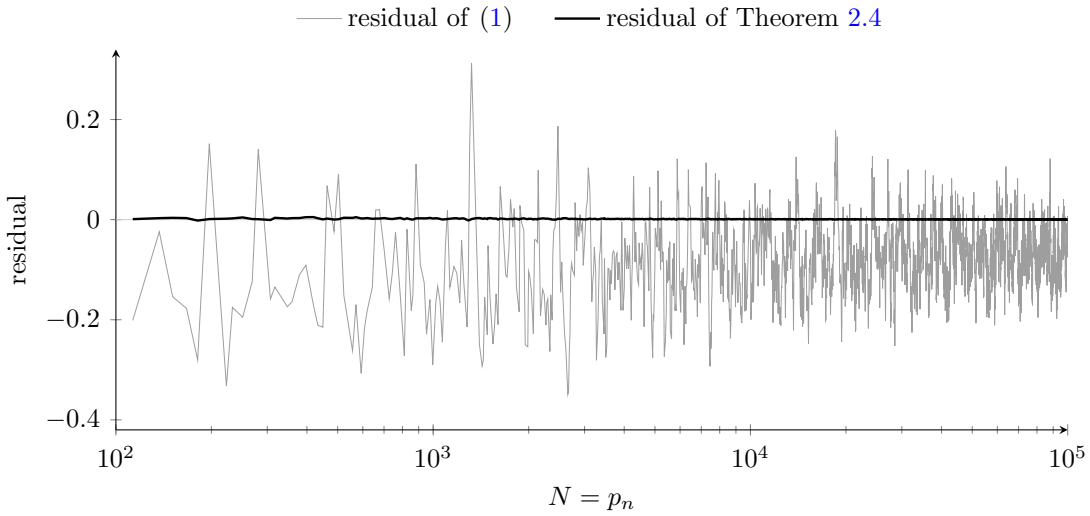


Figure 1: The residual of the heuristic (1) (gray) fluctuates because the density of primes just after q_N fluctuates. Theorem 2.4 accounts for that density through Σ_q, and its residual (black) is below 0.0012 for $N > 20000$. Every third value of n is plotted; all values are in `op3_residuals_full.tsv`.

N	$q - N$	$-\log(I_N - \Lambda)$	residual of (1)	$-\log(t_q \log q\, \Sigma_q)$	residual of Thm. 2.4
1009	4	47.953	−0.252	−0.187	+0.0024
10007	2	130.735	−0.056	+0.006	+0.0015
19609	52	175.991	−0.116	−0.055	+0.0008
31397	72	216.825	−0.083	−0.023	+0.0003
50021	2	266.497	−0.095	−0.036	+0.0003
89689	64	346.084	−0.153	−0.096	+0.0003
99989	2	363.416	−0.010	+0.046	+0.0004

Table 1: Sample values. The fourth column minus the fifth equals the sixth plus the deterministic part of the $o(1)$ in (1), which decays slowly (-0.067 at $N = 1009$, -0.056 at $N = 99989$).

$-\log(t_q \log q\, \Sigma_q)$ for $N > 20000$ gives slope 1.001, intercept -0.058, correlation 0.9998 (0.9995 for $N > 5000$). At the maximal gaps after 19609, 31397 and 89689 the local factor is negative, that is Σ_q is above average: slightly more primes than average follow q. The gap itself lies before q and is absorbed by M_q.

Remark 2.10 (heuristic). The fluctuations of the local factor are a direct measurement of the irregularity of primes at scale $\sqrt{q \log q}$. Over $q \in [70000, 10^5]$ the standard deviation of $\log(t_q \log q\, \Sigma_q)$ is 0.050. The Cramér model predicts $\sqrt{(t_q/2) \log q} \approx 0.10$. A pair-correlation computation with the Hardy–Littlewood singular series predicts ≈ 0.061, and the corresponding conditional mean (+0.0129) matches the observed mean (+0.0128) (`op3_variance.py`).

3 The moment problem

For a weight $w \in L^1(0,1)$, $w \ge 0$, put $F_w(A) = \int_0^1 w(u) P_A(u)\,du$. For an increasing absolutely continuous ϕ with $\phi' = w$, $\mathbb{E}\phi(Z_A) = \phi(0) + F_w(A)$ ([P, Remark 4.4]). Moments correspond to $w(u) = ku^{k-1}$ and truncations to $w\mathbf{1}_{[0,c]}$. Recall $J_s(z) = \int_0^z x^{s-1}(Q(x) - (1-x))\,dx$ and $\alpha_0 = \sup\{b : J_b(1) > 0\} = 2.14312191825192\ldots$. References of the form [P, ...] are to the paper.

Lemma 3.1 (L^1 form of [P, Lemma 3.3])**.** *Let $g \in L^1(0,c)$ with $G(y) = \int_0^y g \ge 0$ for $0 < y \le c$, and let w be nonnegative, continuous and nonincreasing on $[0,c]$. Then $\int_0^c wg = w(c)G(c) + \int_{(0,c]} G\,d(-w) \ge 0$, with strict inequality if $G > 0$ on $(0,c]$ and either $w(c) > 0$ or $d(-w)$ has positive mass on $(0,c)$.*

Proof. Write $w(u) = w(c) + (-w)((u,c])$ and apply Fubini, which is legitimate since $g \in L^1$ and $d(-w)$ is a finite measure: $\int_0^c wg = w(c)G(c) + \int_{(0,c]} \int_0^s g(u)\,du\,d(-w)(s)$. □

3.1 Relative extremality: the exact cone

Theorem 3.2. *Let $w \in L^1(0,1)$, $w \ge 0$. The following are equivalent.*

(a) For every finite $U \subseteq \{2,3,\ldots\}$ and every primitive $A \subseteq U$, $F_w(A) \ge F_w(\mathrm{Min}(U))$.

(b) For every integer $D \ge 2$, $\int_0^1 w(u)\big[Q(u^D) - (1-u^D)\big]du \ge 0$; equivalently $\mathbb{E}\phi(Z_{\mathcal{P}}^{1/D}) \ge \mathbb{E}\phi(V^{1/D})$ with V uniform on $[0,1]$.

Under (b), (a) also holds for every truncation $w\mathbf{1}_{[0,c]}$ and for infinite U and A. If (b) is strict for every D and $w > 0$ almost everywhere, $\mathrm{Min}(U)$ is the unique minimizer.

Here $Z_{\mathcal{P}}^{1/D}$ has the law of $Z_{D\mathcal{P}}$ and $V^{1/D}$ that of $Z_{\{D\}}$. So (b) says that the single compression $D\mathcal{P} \to \{D\}$ does not increase $\mathbb{E}\phi$, and the theorem says that these compressions are the only obstructions.

Proof. (a)⇒(b): take $U = \{D\} \cup D\mathcal{P}_M$, so that $\mathrm{Min}(U) = \{D\}$, and $A = D\mathcal{P}_M$, which is primitive. Then (a) reads $\int_0^1 w(u)P_{\mathcal{P}_M}(u^D)\,du \ge \int_0^1 w(u)(1-u^D)\,du$; let $M \to \infty$ by dominated convergence.

(b)⇒(a): for $d \ge 1$ put $w_d(v) = d^{-1}w(v^{1/d})v^{1/d-1}$. Then $\int_0^y w(u)f(u^d)\,du = \int_0^{y^d} w_d(v)f(v)\,dv$, $(w_d)_e = w_{de}$ and $w_d \in L^1$. Since $g = Q - (1-x)$ changes sign once, from + to − ([P, Remark 4.4]), the running integrals $\int_0^z w_E g$ are smallest at $z = 0$ or $z = 1$. Condition (b) at $D = E$ therefore gives

$$\int_0^z w_E(v)\big[Q(v) - (1-v)\big]dv \;\ge\; 0 \qquad (E \ge 2,\ 0 < z \le 1). \tag{2}$$

We prove by induction on N: *for every $D \ge 1$, $z \in (0,1]$, $U \subseteq [2,N]$ and primitive $A \subseteq U$, $\int_0^z w_D P_A \ge \int_0^z w_D P_{\mathrm{Min}(U)}$.* For $N \le 3$, $A \subseteq U = \mathrm{Min}(U)$. In general, if $A \subseteq \mathrm{Min}(U)$, add the missing elements one at a time; each addition lowers the integral. Otherwise pick $a \in A \backslash \mathrm{Min}(U)$ and a proper divisor $d \in U$ of a. As in the proof of [P, Theorem 4.1], $C_d(A) \subseteq U$ is primitive, and with $B = \{b \in A : d \nmid b\}$ and $T = \{b/d : d \mid b \in A\} \subseteq [2, N/d]$ primitive,

$$\int_0^z w_D(P_A - P_{C_d(A)}) = \int_0^z P_B\,dG, \qquad G(y) = \int_0^{y^d} w_{Dd}(v)\big[P_T(v) - (1-v)\big]dv.$$

Split $P_T - (1-v) = (P_T - P_{\mathcal{P}_{\lfloor N/d \rfloor}}) + (P_{\mathcal{P}_{\lfloor N/d \rfloor}} - Q) + (Q - (1-v))$. The first part has nonnegative running integrals by the induction hypothesis (weight index Dd, $U' = [2, N/d]$, $\mathrm{Min}(U') = \mathcal{P}_{\lfloor N/d \rfloor}$), the second is pointwise ≥ 0, and the third by (2) with $E = Dd \geq 2$. So $G \geq 0$, and Lemma 3.1 with the nonincreasing weight P_B gives $\int_0^z P_B \, dG \geq 0$ (here w_D is only integrable, which is why the L^1 form is needed). The compression lowers the element sum, so iterating ends in a subset of $\mathrm{Min}(U)$. The extensions are as in the proofs of [P, Theorems 1.2 and 4.1]. □

Corollary 3.3. *For $\phi(x) = x^k$, relative extremality holds if and only if $k \leq 2\alpha_0$. For every $k > 2\alpha_0$ the set $2\mathcal{P} = \{4, 6, 10, 14, \dots\}$ beats $\mathrm{Min}(U) = \{2\}$ in $U = \{2, 4, 6, 8, \dots\}$: $\mathbb{E}Z^k_{2\mathcal{P}} < \mathbb{E}Z^k_{\{2\}}$.*

Proof. The D-th condition is $(k/D)J_{k/D}(1) \geq 0$. By variation diminishing, $J_s(1)$ changes sign once, at α_0, so all conditions hold iff $k/2 \leq \alpha_0$. □

So the stopping point $2\alpha_0 \approx 4.286$ of the bootstrap in [P, Remark 4.4] is not a weakness of that method: no argument that uses only the relative structure can pass it.

Remark 3.4 (when $D = 2$ suffices). If $u \mapsto \log(1/u)\,(uw(u))'/w(u)$ is nonincreasing on $(0, 1)$ (for instance $w = ku^{k-1}$), then $w_{D'}/w_D$ is nonincreasing for $D' > D$, and the single condition $D = 2$ implies all others: $\int w_{D'} g \geq (w_{D'}/w_D)(x^*) \int w_D g$, where x^* is the sign change of g.

3.2 Absolute extremality: ordered compressions

The obstruction of Corollary 3.3 is a compression at 2 with a prime-like quotient block. In the absolute problem it can be avoided.

Theorem 3.5. *If $J_b(1) > 0$, then for every $0 < k \leq 3b$ and $c \in (0, 1]$ the primes uniquely minimize $\mathbb{E}\min(Z^k_A, c^k)$, over primitive $A \subseteq \{2, \dots, N\}$ for every N and over all primitive A. In particular this holds for $0 < k \leq 51/8$ ($b = 17/8$) and for every $k < 3\alpha_0 = 6.4293\dots$.*

Proof. For $k \leq 1$ see [P, Remark 4.4]. Let $1 < k \leq 3b$. Descend from A as follows. *While some composite of A has an odd prime factor p, compress at p.* The block T is an arbitrary primitive set. If $p \geq k$, [P, Lemma 3.1] applies. If $p < k$, the running integral of $ku^{k-1}E_T(u^p)$ splits as in [P, Remark 4.4] into the comparison of T with $\mathcal{P}$ at exponent $s = k/p \leq k/3 \leq b$, which is nonnegative by the moment theorem at smaller exponents, plus $J_s(y^p) > 0$. Once no composite has an odd prime factor, every composite is a power of 2, and a primitive set contains at most one, say 2^j with $j \geq 2$. *Compress at 2:* the only even element is 2^j, so $T = \{2^{j-1}\}$ and $E_T(v) = v - v^{2^{j-1}} > 0$ on $(0, 1)$. This step is free. Finally add the missing primes. Infinite sets are treated as in the proof of [P, Theorem 1.2], choosing the compression prime by the same rule. The comparison at smaller exponents needs minimality at exponents in $[1, b]$, which is known: $J_b(1) > 0$ forces $b < \alpha_0 < 17/4$, and [P, Remark 4.4] covers $[1, 17/4]$. □

This is formalized in `RntPaper/MomentsOdd.lean` (Lean 4.35, built against the paper's pinned Mathlib and LeanCert): `moment_primes_lt_odd`, `moment_primesLE_lt_odd` ($0 < k \leq 51/8$) and `moment_primes_lt_odd_of_Jw` ($0 < k \leq 3b$ whenever $J_b(1) > 0$). All three depend only on `propext`, `Classical.choice` and `Quot.sound`.

Treating the primes 3 and 5 by envelopes instead of J pushes further.

Lemma 3.6 (envelopes)**.** *Let T be an antichain in $\langle 2,3\rangle \setminus \{1\}$. If $2 \in T$ or $3 \in T$, then $P_T \geq (1-v^2)(1-v^3)$ on $[0,1]$; otherwise $P_T \geq H_{23} := \prod_{n\in\langle 2,3\rangle,\, n\geq 4}(1-v^n)$. Let T be an antichain in $\langle 2,3,5\rangle \setminus \{1\}$ and $R_0 = T \cap \{2,3,5\}$. Then $P_T \geq E_{R_0}$, where $E_{\{2,3,5\}} = (1-v^2)(1-v^3)(1-v^5)$, $E_{\{2,3\}} = (1-v^2)(1-v^3)(1-v^{25})$, $E_{\{2,5\}} = (1-v^2)(1-v^5)(1-v^9)$, $E_{\{3,5\}} = (1-v^3)(1-v^4)(1-v^5)$, and for $|R_0| \leq 1$ $E_{R_0} = \prod_{r\in R_0}(1-v^r)\prod_n(1-v^n)$ over $n \in \langle 2,3,5\rangle$, $n \geq 4$, $n \notin \{2,3,5\}$, divisible by no element of R_0.*

Proof. If $2 \in T$, the other elements are odd, hence powers of 3, and pairwise comparable, so there is at most one. Similarly if $3 \in T$. If $\{2,3\} \subseteq T$ in the second case, the other elements are powers of 5, at least 25, at most one; the other pairs are alike. In all cases P_T is at least the product over the largest set of exponents T can contain. □

Theorem 3.7. *For every $0 < k < 3 + 6\sqrt{3} = 13.3923\ldots$ and $c \in (0,1]$ the primes uniquely minimize $\mathbb{E}\min(Z_A^k, c^k)$ over primitive sets (finite in $\{2,\ldots,N\}$ for every N, and arbitrary).*

Proof. For $k \leq 51/8$ this is Theorem 3.5; let $51/8 < k < 3+6\sqrt{3}$. Order the compressions: (i) at primes $p \geq 7$ dividing a composite, as in Theorem 3.5, which needs $k/7 \leq 17/8$; (ii) at 5, once every composite is 5-smooth; (iii) at 3, once every composite is 3-smooth; (iv) at 2, with a singleton block. In step (ii) the block T lies in $\langle 2,3,5\rangle$, in step (iii) in $\langle 2,3\rangle$. By Lemma 3.6 and Lemma 3.1 it suffices that, for each envelope E and $s = k/p$, the running integrals $R_s(z) = \int_0^z v^{s-1}(E(v)-(1-v))\,dv$ are positive on $(0,1]$. The script `envelope_certificate.py` verifies this in exact rational arithmetic, with one of two certificates:

(SC) E is a polynomial and $E-(1-v) = v(1-v)h(v)$ with exactly one root of h in $(0,1)$ and $h(0) > 0 > h(1)$ (Sturm). Then $R_s(z) \geq \min(R_s(0^+), R_s(1))$, and $R_s(1) > 0$ implies $R_{s'}(1) > 0$ for $s' \leq s$.

(V1) A polynomial $L \leq E$ on $[0, v_1]$ with $L-(1-v) \geq 0$ there, and $\int_0^{v_1} v^{s-1}(L-(1-v)) > \int_{v_1}^1 v^{s-1}(1-v)$. Then, since $E \geq 0$, $R_s > 0$ on $(0,1]$, and the inequality persists for $s' \leq s$.

For $(1-v^2)(1-v^3)$ the computation is exact: $R_s(1) = \frac{11+2s-s^2}{(s+1)(s+2)(s+3)(s+5)}$, which is positive iff $s < 1+2\sqrt{3}$. This is the bottleneck, at $k = 3s$. The other certificates are: H_{23} by (V1) at $s = 5$; $E_{\{2,3,5\}}$ by (SC) at $s = 75/28 > (3+6\sqrt{3})/5$, where $R_s(1) = 9.69 \cdot 10^{-5}$; the other three polynomial envelopes by (SC) at $s = 3$; the four infinite envelopes by (V1) at $s = 3$. Step (i) needs $(3+6\sqrt{3})/7 = 1.913 < 17/8$. □

The configuration that blocks this method is $A \supseteq \{6,9\}$ with $2,3 \notin A$, compressed at 3 with $T = \{2,3\}$. It is exactly the configuration of A_9.

3.3 The exact threshold

Theorem 3.8. *Let $k_9 = 125.84045406944565748\ldots$ be the unique positive root of $\mathbb{E}Z^k_{\{4,5,6,7,9\}} = \mathbb{E}Z^k_{\{2,3,5,7\}}$. For every $0 < k \leq k_9$, every $N \geq 2$, every primitive $A \subseteq \{2,\ldots,N\}$ and every $c \in (0,1]$,*

$$\mathbb{E}\min(Z^k_{\mathcal{P}_N}, c^k) \;\leq\; \mathbb{E}\min(Z^k_A, c^k),$$

with strict inequality if $k < k_9$ and $A \neq \mathcal{P}_N$. The same holds with $\mathcal{P}$ in place of $\mathcal{P}_N$ for every primitive A, finite or infinite. For every $k > k_9$ the inequality fails at $N = 9$, $A = A_9 = \{4,5,6,7,9\}$, $c = 1$. Hence the largest k for which $\mathcal{P}_N$ minimizes $\mathbb{E}Z^k_A$ over primitive $A \subseteq \{2,\ldots,N\}$ for every N is exactly k_9.

The proof is a weighted version of the ordered descent of Theorem 3.7. Two changes make it reach k_9: the factor contributed by the large primes is kept in every comparison, and the primes $2, 3, 5, 7$ are not compressed one at a time but replaced together.

Lemma 3.9 (weighted exchange)**.** *Let p, B, T be as in [P, Proposition 3.4], $B_0 \subseteq B$, $k > 0$ and $c \in (0, 1]$. If*

$$G(y) = \int_0^y k u^{k-1} P_{B_0}(u) \big[P_T(u^p) - (1 - u^p)\big]\, du \;\geq\; 0 \qquad (0 < y \leq c),$$

then $\mathbb{E}\min(Z^k_{B\cup\{p\}}, c^k) \leq \mathbb{E}\min(Z^k_{B\cup pT}, c^k)$, with strict inequality if $G > 0$ on $(0, c]$.

Proof. The difference of the two sides is $\int_0^c P_{B\setminus B_0}\, dG$, and $w = P_{B\setminus B_0}$ is continuous, nonincreasing and nonnegative; apply Lemma 3.1. For strictness: if $B \setminus B_0 = \varnothing$ the difference is $G(c)$; otherwise, if $c < 1$ then $w(c) > 0$, and if $c = 1$ then $w(1) = 0$ and $d(-w)$ has mass 1 on $(0, 1)$. □

Proof of Theorem 3.8. Fix $0 < k \leq k_9$, $c \in (0, 1]$, N and a primitive $A \subseteq [2, N]$. Write $\mathcal{M}(A) = \mathbb{E}\min(Z^k_A, c^k)$. We transform A into $\mathcal{P}_N$ by steps that do not increase $\mathcal{M}$; adding a prime that divides no element of A always lowers $\mathcal{M}$.

(i) While some composite of A has a prime factor $p \geq 61$, compress at such a p. This is admissible as in Theorem 3.5: either $p \geq k$ ([P, Lemma 3.1]), or $1 \leq k/p \leq k_9/61 < 2.07 < 17/8$, and minimality at exponent k/p is [P, Remark 4.4].

(ii) For $p = 59, 53, \ldots, 13, 11$ in turn: every composite is now p-smooth, so no composite element of A is divisible by a prime $q \in (p, N]$, and we add every such q missing from A. If some composite is divisible by p, compress at p. The block $T = \{a/p : p \mid a \in A\}$ is an antichain of p-smooth numbers ≥ 2, and since $p \max T \leq N$ the remaining elements contain $B_0 = \mathcal{P} \cap (p, p \max T]$. By Lemma 3.9 this step is admissible if

$$\int_0^y k u^{k-1} \prod_{q\in Q} (1 - u^q) \left[P_T(u^p) - (1 - u^p)\right] du \;\geq\; 0 \qquad (0 < y \leq 1) \tag{3}$$

for some set Q of primes in $(p, p \max T]$.

(iii) Now every composite is 7-smooth. Add every missing prime in $[11, N]$ (none divides an element of A; this is needed for 11, which step (ii) never adds), so that $A \supseteq \mathcal{P} \cap [11, N]$, and add each prime $r \leq 7$, $r \leq N$, that divides no element of A. If $N \leq 6$, compare A with $\mathcal{P}_N$ directly. Otherwise $A = (\mathcal{P} \cap [11, N]) \cup S$, where S is an antichain of 7-smooth numbers in which each of $2, 3, 5, 7$ divides some element, and

$$\mathcal{M}(A) - \mathcal{M}(\mathcal{P}_N) = \int_0^c k u^{k-1} P_{\mathcal{P}\cap[11,N]}(u) \left[P_S(u) - P_{\{2,3,5,7\}}(u)\right] du.$$

Since $N \geq \max S$, Lemma 3.1 reduces nonnegativity to

$$\int_0^y k u^{k-1} \prod_{11\leq q\leq \max S} (1 - u^q) \left[P_S(u) - P_{\{2,3,5,7\}}(u)\right] du \;\geq\; 0 \qquad (0 < y \leq 1). \tag{4}$$

Proposition 3.10 supplies (3), (4) and the case $N \leq 6$. For $k < k_9$ every certificate gives strict positivity on $(0, 1]$, so any nontrivial step is strict. For infinite A, the inequality follows

from the finite one by truncation and dominated convergence. For strictness, either some composite has a prime factor ≥ 61, and one strict compression followed by the lower bound suffices, or all composites are 59-smooth. In that case they form an antichain in $\mathbb{N}^{17}$, which is finite by Dickson's lemma. Add all primes dividing no composite, and run steps (ii)–(iii) on the resulting set: B then contains infinitely many primes, which is harmless since P_B is still continuous and nonincreasing on $[0,1]$, and the finitely many compressions end at $\mathcal{P}$ with at least one strict step. At $k = k_9$ every certificate except that of A_9 (and of its subsets, which are dominated pointwise) is strict, since they hold up to 126; the final step with $S = A_9$ is strict unless $N \leq 10$ (so that no prime ≥ 11 enters the weight) and $c = 1$ (otherwise $R_{k_9}(c) > 0$). So at $k = k_9$ equality holds only for $A = A_9$, $N \in \{9, 10\}$, $c = 1$. Finally, $P_{A_9} - P_{\mathcal{P}_9}$ changes sign exactly once on $(0,1)$, from $+$ to $-$. By variation diminishing, $k \mapsto \mathbb{E}Z_{A_9}^k - \mathbb{E}Z_{\mathcal{P}_9}^k$ changes sign exactly once, at k_9, and is negative for $k > k_9$. □

Proposition 3.10 (computer-verified)**.** *(a) For every prime $11 \leq p \leq 59$ and every antichain T of p-smooth numbers ≥ 2, condition (3) holds for every $0 < k \leq 126$.*

(b) For every antichain S of 7-smooth numbers in which each of $2, 3, 5, 7$ divides some element, condition (4) holds for every $0 < k \leq 126$ if $S \not\subseteq A_9$, and for every $0 < k \leq k_9$ if $S \subseteq A_9$. (The proper subsets of A_9 with this covering property and maximum 9, namely $\{4, 5, 7, 9\}$ and $\{5, 6, 7, 9\}$, even satisfy $P_S > P_{\{2,3,5,7\}}$ on $(0,1)$; this is checked exactly, but not needed.)

(c) For $N \leq 6$ and every primitive $A \subseteq [2, N]$, $\int_0^y ku^{k-1}(P_A - P_{\mathcal{P}_N})\,du \geq 0$ for all $y \in (0,1]$ and $0 < k \leq 126$.

Verification. First, finitely many cases.

- *Maximality.* If $T \subseteq T'$ and $\max T' = \max T$, the integrand of (3) for T dominates the one for T' pointwise, since the weight is the same and $P_T \geq P_{T'}$; likewise for S. So it suffices to treat T maximal among antichains of p-smooth numbers in $[2, \max T]$, and S maximal among antichains of 7-smooth numbers in $[2, \max S]$.

- *Explicit cases.* All such T with $\max T < m_p$, for $(p, m_p) = (11, 30)$, $(13, 24)$, $(17, 20)$, $(19, 16)$, $(23, 14)$, $(29, 12)$, $(31, 10)$, $(37, 9)$, $(41, 8)$, $(43, 8)$, $(47, 6)$, $(53, 6)$, $(59, 5)$, and all such S with $\max S < 64$, are enumerated: 928 blocks and 3113 sets.

- *Tails of (a).* If $\max T \geq m_p$, take $Q \subseteq \mathcal{P} \cap (p, pm_p]$ and cut T at $B_p \leq m_p$ ($B_p = 24, 20, 16, 14, 12, 10, 9, 8, 7, 7, 6, 6, 5$). The part below B_p lies in a maximal antichain M of p-smooth numbers below B_p, so $P_T(v) \geq P_M(v) \prod_{n \geq B_p}(1 - v^n)$. This gives 243 envelope conditions.

- *Tails of (b).* If $\max S \geq 64$, write $S = R_0 \cup S'$ with $R_0 = S \cap \{2, 3, 5, 7\}$. Then S' consists of composites of $\langle R' \rangle$, where $R' = \{2, 3, 5, 7\} \setminus R_0$, and $P_S - P_{\{2,3,5,7\}} = P_{R_0}(P_{S'} - P_{R'})$. Put $A_0 = S' \cap [4, 24)$, an arbitrary antichain. The elements of S' that are ≥ 24 are composites of $\langle R' \rangle$ divisible by no element of A_0. If none of these is ≥ 64, the case is vacuous. Otherwise $P_{S'} \geq P_{A_0} \prod_{n \in C(A_0)}(1 - u^n)$, where $C(A_0)$ is that set of composites. With the weight $P_{R_0} \prod_{11 \leq q \leq 64}(1 - u^q)$ this gives 1031 envelope conditions, 12 of them vacuous (the Lean proof certifies these 12 by the same envelope test as the others). (The full weight in (4) contains this one; the extra factor is nonincreasing, so Lemma 3.1 transfers the condition.)

- *Small N*: 35 nonempty primitive sets (the Lean enumeration also checks $A = \varnothing$ for each N, 40 lists in all).

Each case is then verified in exact integer and rational arithmetic (`stage1/cert_lib.py`, `stage1/cert_stage3.py`).

(SC) For a polynomial integrand $W \cdot D$ with $W > 0$: exact real-root isolation shows that D is positive on $(0, 1)$ or has one sign change there, from $+$ to $-$. In the latter case $R_{126}(1) = \sum_n g_n \, 126/(126 + n)$ is computed exactly and is positive. By variation diminishing the running-integral condition then holds for all $0 < k \le 126$.

(V1) For an envelope, the certificate of Theorem 3.7, now with the weight: a polynomial lower bound L on $[0, u_1]$; nonnegativity of L minus the target on $[0, u_1]$, by the Weierstrass and Bonferroni inequalities near 0 and a monotone partition elsewhere; and an exact comparison of the two integrals. For the tails of (b), L keeps the factors $1 - u^n$ with $n \in A_0$ or $n \in C(A_0)$, $n < N_2 = 120$, and bounds the product over *all* $n \ge N_2$ (a superset of $C(A_0) \cap [N_2, \infty)$) by $1 - u^{N_2}/(1 - u_1)$. For the tails of (a), the cut is at B_p in the variable $v = u^p$, with $\prod_{n \ge B_p}(1 - v^n) \ge 1 - v^{B_p}/(1 - v_1)$.

(A_9) One sign change, and the exact values $R_{125.84}(1) > 0 > R_{125.85}(1)$, which bracket k_9. By variation diminishing the condition holds exactly for $k \le k_9$.

All 5350 certificates pass. Apart from A_9, the smallest margins are $R_{126}(1) = 5.8 \cdot 10^{-8}$ for the sets S and $1.4 \cdot 10^{-5}$ for the blocks T. A random sample of the exact values was recomputed by 40-digit quadrature from independently built integrands, and agrees to relative precision 10^{-16}. □

Remark 3.11 (why $2, 3, 5, 7$ together). With the weights, single-prime compressions at 5 and 7 fail before k_9: at about 54.6 and 87.0 respectively, both for the block $T = \{2, 3, 5\}$. At 11 and beyond the margins are large; the smallest explicit threshold is 198.6, at $p = 11$. The joint comparison of the 7-smooth part is tight at A_9 and nowhere else. The next smallest threshold is 139.9, for $S = \{4, 6, 7, 9, 10, 15, 25\}$, the set behind $N = 25$ below.

Remark 3.12 (consistency checks). The theorem agrees with direct computation. Over all maximal primitive $A \subseteq \{2, \dots, N\}$ with $N \le 46$ (22,299 sets at $N = 46$), the first exponent with $\mathbb{E}Z_A^k < \mathbb{E}Z_{\mathcal{P}_N}^k$ is smallest at $N = 9, 10$ (125.84, from A_9); then come 149.0 ($N = 11, 12$), 171.3, 141.2 ($N = 15$), …, 139.92 ($N = 25$, adding 25), …, and 171.0 at $N = 46$. About 27,000 further competitors $(\mathcal{P}_N \setminus R) \cup S$ were tested at $c = 1$, with $R \subseteq \{2, 3, 5, 7, 11\}$ and elements up to 3000; none crosses below k_9. An independent reviewer (`review2/`) enumerated all 2,194,792 maximal primitive subsets of $\{2, \dots, N\}$ for $7 \le N \le 62$, at all truncation levels c, and ran random and simulated-annealing searches up to $N = 400$: nothing falls below k_9, and only A_9 and the set at $N = 25$ fall below 140. The same reviewer re-verified every case of Proposition 3.10 by different exact methods (independent root isolation, a direct exact proof that $R_{126}(y) > 0$ for all y, and outward-rounded interval arithmetic for the tails). At $k = k_9$ equality holds only for $A = A_9$ with $N \in \{9, 10\}$ and $c = 1$. The Lean development already contains the case $N = 9$ for $1 \le k \le 3146/25$ (`primes_minimize_Mk_N9`); for $0 < k < 1$ the case $N = 9$ follows from the concave case.

4 Lean formalization

The following are formalized in `lean/RntPaper/` (Lean 4.35, the paper's pinned Mathlib and LeanCert), with no `sorry`. Theorem 3.2 and Corollary 3.3: `Relative.lean` (`relative_cone_iff`, `relative_cone_le`, `relative_cone_lt`, `relative_powers_iff`, `twoPrimes_moment_lt`), for every weight $w \in L^1(0,1)$, $w \geq 0$. Theorem 3.5: `MomentsOdd.lean`. Theorem 3.7: `Moments1339.lean`. These use only the axioms `propext`, `Classical.choice`, `Quot.sound`.

Theorem 3.8: `K9Theorem.lean` (`moment_primesLE_k9`, `moment_primes_k9`, `moment_A9_lt`, `k9_threshold`, `k9_unique_root`), with k_9 defined as the root of $\mathbb{E}Z^k_{A_9} = \mathbb{E}Z^k_{\mathcal{P}_9}$ in $(125.84, 125.85)$. Each executable check of Proposition 3.10 has a soundness theorem (single sign change with variation diminishing, and the envelope test), and the completeness of the case enumeration is proved; the envelope parameters are re-verified inputs. The checks themselves run by `native_decide`, so this theorem additionally trusts Lean's compiler, as do the two `native_decide` computations of [P]. Not formalized: the equality case at $k = k_9$ and the remark on the proper subsets of A_9. An independent audit is in `supplement/discovery/review3/`.

Files

- `supplement/discovery/op3_local.py`: telescoping terms, saddle data, the predictions of §2; `op3_tables.py`, `op3_variance.py`.
- `supplement/discovery/op2_relative.py`: α_0 and the relative test. `op2_absolute_scan.py`, `op2_scan_fast.py`, `op2_families.py`, `stage1/cex_search.py`: the consistency checks for Theorem 3.8.
- `supplement/discovery/envelope_certificate.py`: exact certificates for Theorem 3.7.
- `supplement/discovery/stage1/`: `cert_lib.py` and `cert_stage3.py` (exact certificates for Proposition 3.10; logs `cert_part1_log.jsonl`, `cert_part2_log.jsonl`), `cross_check.py` (independent quadrature), and the floating-point explorations `explore.py`, `stage3_explore.py`, `merged_tail2.py`.
- `lean/RntPaper/`: the formalization (§4).
- `supplement/discovery/review/`: an independent review of §§2–3.1 with its own recomputations.

Floating-point computations are evidence, not certificates; the envelope certificates, the certificates of Proposition 3.10 and the Lean proofs are exact.

References

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[2] H. Cramér, Some theorems concerning prime numbers, *Ark. Mat. Astr. Fys.* **15** (1920), no. 5, 1–33.

[3] M. N. Huxley, On the difference between consecutive primes, *Invent. Math.* **15** (1972), 164–170.

[4] A. Kumchev, *The distribution of prime numbers*, lecture notes (2005), §5.6, Theorem 7, https://tigerweb.towson.edu/akumchev/distributionofprimesnotes.pdf.

[5] H. L. Montgomery, *Topics in Multiplicative Number Theory*, Lecture Notes in Math. 227, Springer, 1971.

[6] H. L. Montgomery and R. C. Vaughan, The large sieve, *Mathematika* **20** (1973), 119–134.